THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The roots of $x^2 + 4x - 12 = 0$ differ by: 1

(a) 8 **(b)** 12 **(c)** 4 **(d)** 9

2. If $x = 6$ is a root of $x^2 - 13x + k = 0$, then $k =$ 1

(a) 7 **(b)** -42 **(c)** 42 **(d)** 114

3. The equation $x^2 + 7x + 10 = 0$ has: 1

(a) no real roots **(b)** three real roots

(c) two distinct real roots **(d)** two equal real roots

4. The larger root of $x^2 - 13x + 42 = 0$ is: 1

(a) 7 **(b)** 6 **(c)** 13 **(d)** 42

5. The sum of the roots of $4x^2 + 5x - 6 = 0$ is: 1

(a) $-\frac{5}{4}$ **(b)** $\frac{3}{2}$ **(c)** $\frac{5}{4}$ **(d)** $-\frac{3}{2}$

6. The product of the roots of $4x^2 + 5x - 6 = 0$ is: 1

(a) $\frac{3}{2}$ **(b)** $-\frac{5}{4}$ **(c)** $-\frac{3}{2}$ **(d)** $\frac{5}{4}$

7. The roots of $x^2 - 9x + 8 = 0$ differ by: 1

(a) 9 **(b)** 7 **(c)** 14 **(d)** 8

8. If $x = 7$ is a root of $x^2 - 12x + k = 0$, then $k =$ 1

(a) 133 **(b)** -35 **(c)** 35 **(d)** 5

9. The equation $x^2 + 2x + 9 = 0$ has: 1

(a) no real roots **(b)** two equal real roots

(c) three real roots **(d)** two distinct real roots

10. The larger root of $x^2 - 15x + 26 = 0$ is: 1

(a) 15 **(b)** 2 **(c)** 13 **(d)** 26

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find k so that $kx^2 - 12x + 3 = 0$ has equal roots. 2

12. Solve for x: $3x^2 - 19x + 6 = 0$ by the quadratic formula. 2

13. If $x = 6$ is a root of $x^2 - 13x + k = 0$, find k and the other root. 2

14. Find k so that $kx^2 - 10x + 5 = 0$ has equal roots. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve for x: $\frac{1}{x} + \frac{1}{x+2} = \frac{5}{12}$. 3

16. Find the nature of the roots of $5x^2 + 6x + 1 = 0$ and, if real, find them. 3

17. Find the roots of $x^2 + 4x - 12 = 0$ using the quadratic formula (if they exist). 3

OR

Solve for x: $\frac{1}{x} + \frac{1}{x+1} = \frac{11}{30}$. 3

18. Find the nature of the roots of $2x^2 - 3x + 4 = 0$ and, if real, find them. 3

OR

Find the roots of $3x^2 - 14x + 8 = 0$ using the quadratic formula (if they exist). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The altitude of a right triangle is 7 cm less than its base and the hypotenuse is 13 cm. Find the base and altitude. 5

OR

The sum of the ages of two friends is 36 years. 4 years ago the product of their ages was 147. Find their present ages. 5

20. The altitude of a right triangle is 17 cm less than its base and the hypotenuse is 25 cm. Find the base and altitude. 5

OR

The sum of the ages of two friends is 46 years. 8 years ago the product of their ages was 144. Find their present ages. 5

END OF QUESTION PAPER — Total: 40 Marks

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